

Stoichiometry

Mole ratios and mass calculations from balanced equations

Name _____ Class _____ Date _____

DATA

H = 1.008 g/mol C = 12.01 g/mol O = 16.00 g/mol Mg = 24.31 g/mol Cl = 35.45 g/mol

K = 39.10 g/mol Fe = 55.85 g/mol

Road map: grams → moles → (mole ratio) → moles → grams. Check that each equation is balanced.

A. Mole ratios

1. For $4 \text{ Al} + 3 \text{ O}_2 \longrightarrow 2 \text{ Al}_2\text{O}_3$, write the mole ratio of (a) O_2 to Al and (b) Al_2O_3 to Al.

2. Using the same equation, how many moles of O_2 react with 6.0 mol of aluminium?

B. Mass calculations

3. Magnesium burns: $2 \text{ Mg} + \text{O}_2 \longrightarrow 2 \text{ MgO}$. What mass of magnesium oxide forms from 12.15 g of magnesium?

4. Potassium chlorate decomposes on heating: $2 \text{ KClO}_3 \longrightarrow 2 \text{ KCl} + 3 \text{ O}_2$. What mass of oxygen forms from 24.5 g of KClO_3 ?

5. Methane burns: $\text{CH}_4 + 2 \text{O}_2 \longrightarrow \text{CO}_2 + 2 \text{H}_2\text{O}$. What mass of carbon dioxide forms from 8.02 g of methane?

6. Propane burns: $\text{C}_3\text{H}_8 + 5 \text{O}_2 \longrightarrow 3 \text{CO}_2 + 4 \text{H}_2\text{O}$. What mass of water forms from 11.0 g of propane?

C. Challenge

7. Iron is made from iron(III) oxide and carbon monoxide: $\text{Fe}_2\text{O}_3 + \text{CO} \longrightarrow \text{Fe} + \text{CO}_2$ (unbalanced).

(a) Balance the equation. (b) What mass of iron can be made from 100.0 g of Fe_2O_3 ?

8. A student says: "In $2 \text{Mg} + \text{O}_2 \longrightarrow 2 \text{MgO}$, 2 g of magnesium reacts with 1 g of oxygen." Explain the mistake.